EPG-NSGA-II

Engineering-Prior-Guided NSGA-II for Differentiated Capacity-Load Ratio Optimization in County-Level 110 kV Grids with High Distributed PV Penetration

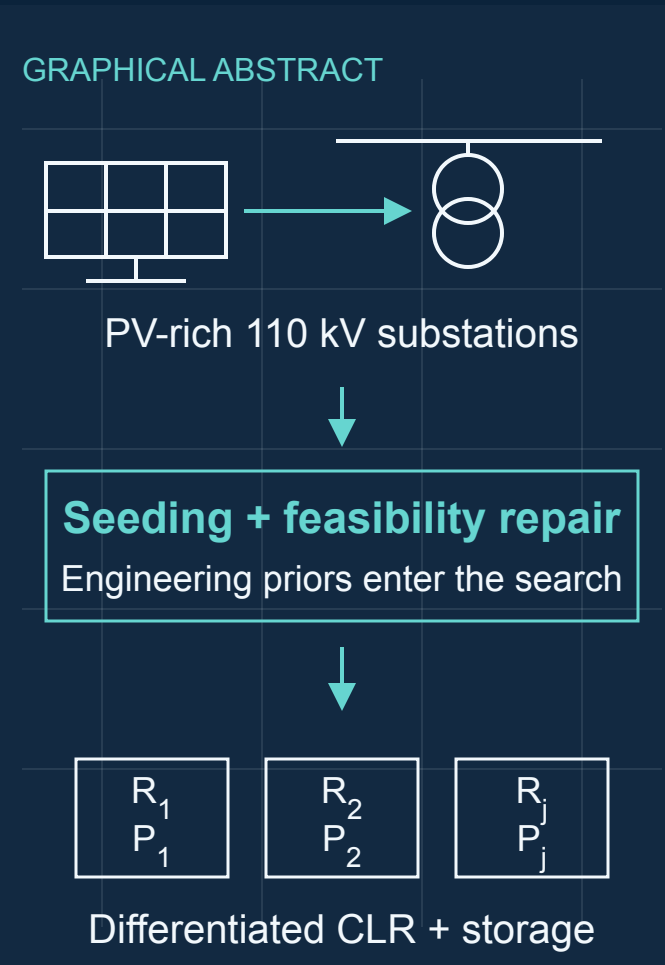
Dunbin Xu · Ke Wang · Yan Xu* · Ziyue Huang · Fei Geng · Rui Yuan

State Grid Xuzhou Power Supply Company

State Grid Jiangsu Electric Power Design & Consulting Co., Ltd.
Xuzhou Survey and Design Branch

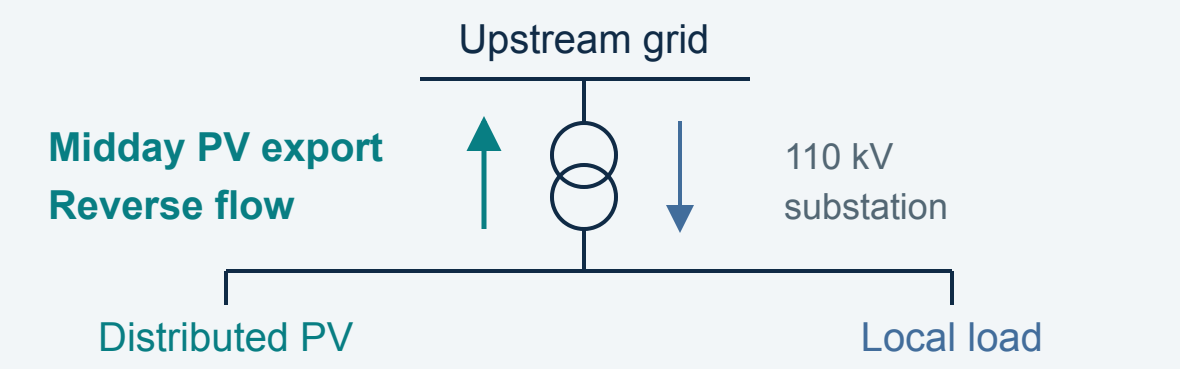
* Corresponding author: Yan Xu

GRAPHICAL ABSTRACT



High PV calls for differentiated planning

a Reverse-flow coupling and planning model



Decisions: **CLR R_j** and **storage power P_j (MW)**

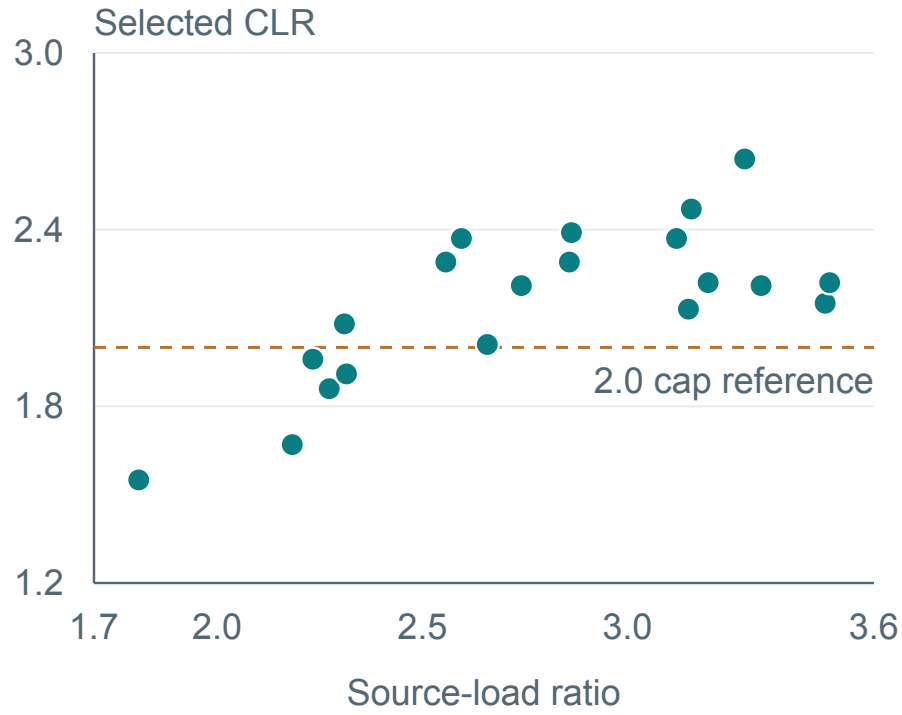
CLR = transformer capacity / forecast peak load

$$1.8 \le \frac{\sum_{j=1}^{K} R_j L_j^p}{\sum_{j=1}^{K} L_j^p} \le 3.0$$

Minimize annualized cost and EENS

million CNY/yr · MWh/yr · N−1 supply inadequacy

b Station heterogeneity | K = 20



20 stations · Selected CLR 1.55–2.64

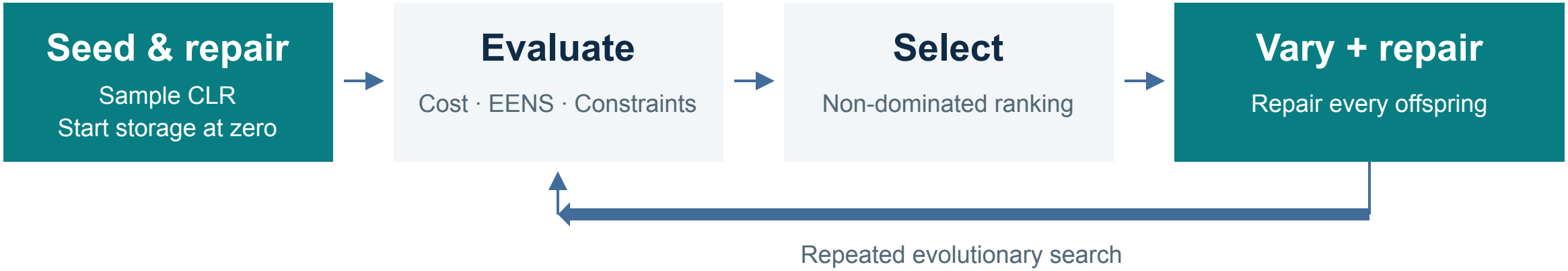
Source-load ratio = installed PV capacity / current peak load

Constraints: station reverse flow · aggregate CLR band · optional upstream reverse-flow limit

Source: camera-ready paper, Sections I–II and Fig. 1. Public-data benchmark; all 20 station records shown.

Engineering priors guide the search

Standard NSGA-II + **engineering-prior seeding** + **feasibility repair**



ENGINEERING-PRIOR REPAIR

1 Aggregate CLR

Bounded common offset

2 Station reverse flow

Increase storage power

3 Upstream limit

Greedy allocation when enabled

β = 0.85

$$P_j \leftarrow \mathrm{clip}\left(\max\left\{P_j, \left[r_j^0 - \beta R_j L_j^p\right]^+\right\}, 0, P_{\max}\right)$$

Residual violations: constraint domination. Standard variation and environmental selection are retained.

Source: Section II-B, Algorithm 1. r0: pre-storage reverse injection; Lp: forecast peak load; Pmax: storage rating bound.

Better search, lower modeled planning cost

Mean final HV | K = 40

+4.8%

EPG: 400 gen · NSGA-II: 800 gen

Annualized cost | K = 20

−3.4%

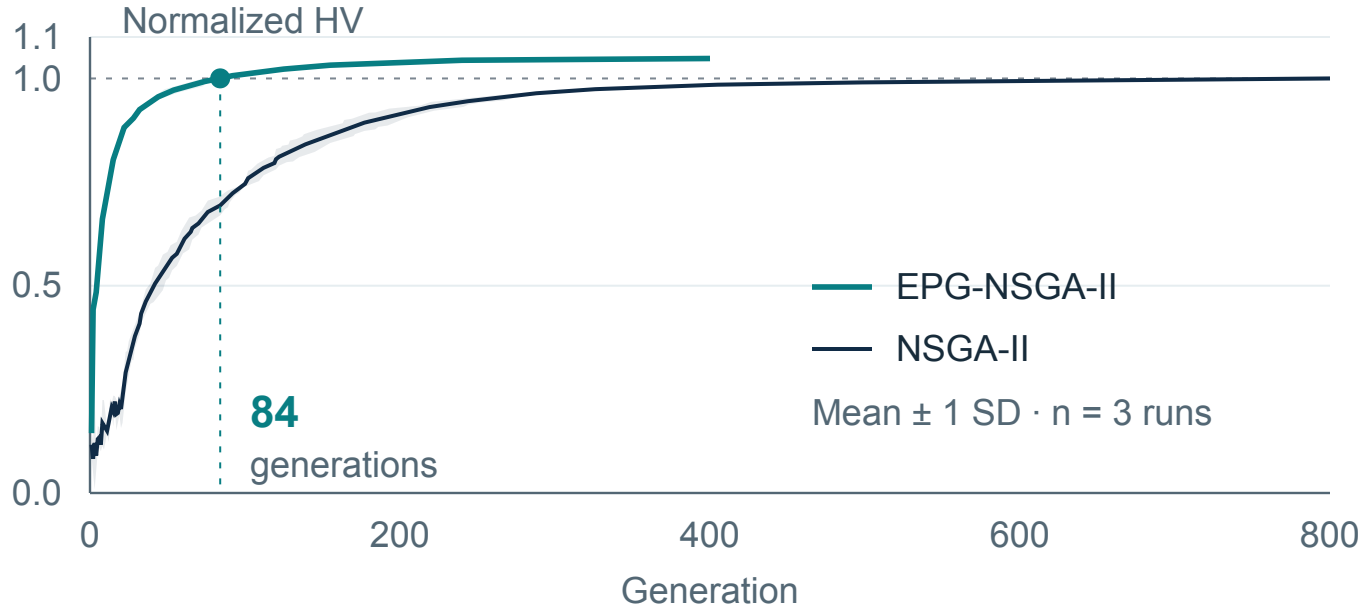
236.40 → 228.43 million CNY/yr

Modeled EENS | K = 20

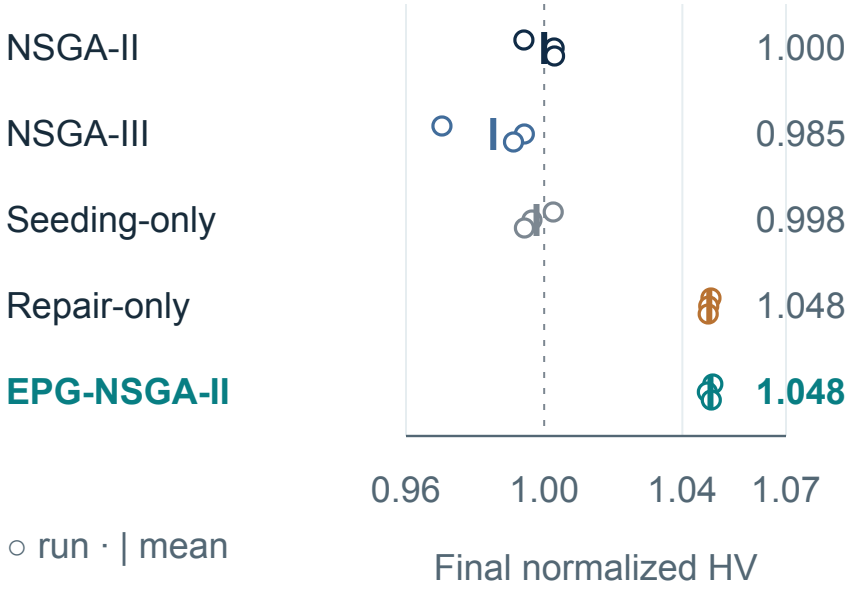
0.00

561.00 → 0.00 MWh/yr

a Search dynamics | K = 40



b Ablation | K = 40

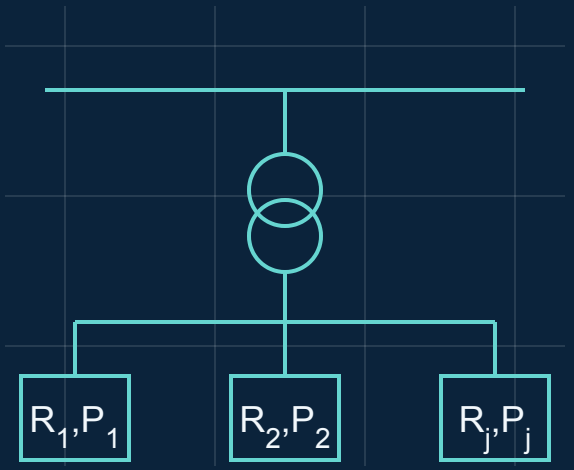


Population 100 · Three independent runs · HV normalized to NSGA-II final mean · Common reference point

Table I / Section III. Public-data model; storage energy/SOC limits and real-grid calibration remain future work.

Thank you

Questions & discussion



Engineering priors support **more effective search** and differentiated CLR and storage planning.

ENGINEERING KNOWLEDGE → CANDIDATE CONSTRUCTION → PLANNING TRADE-OFFS